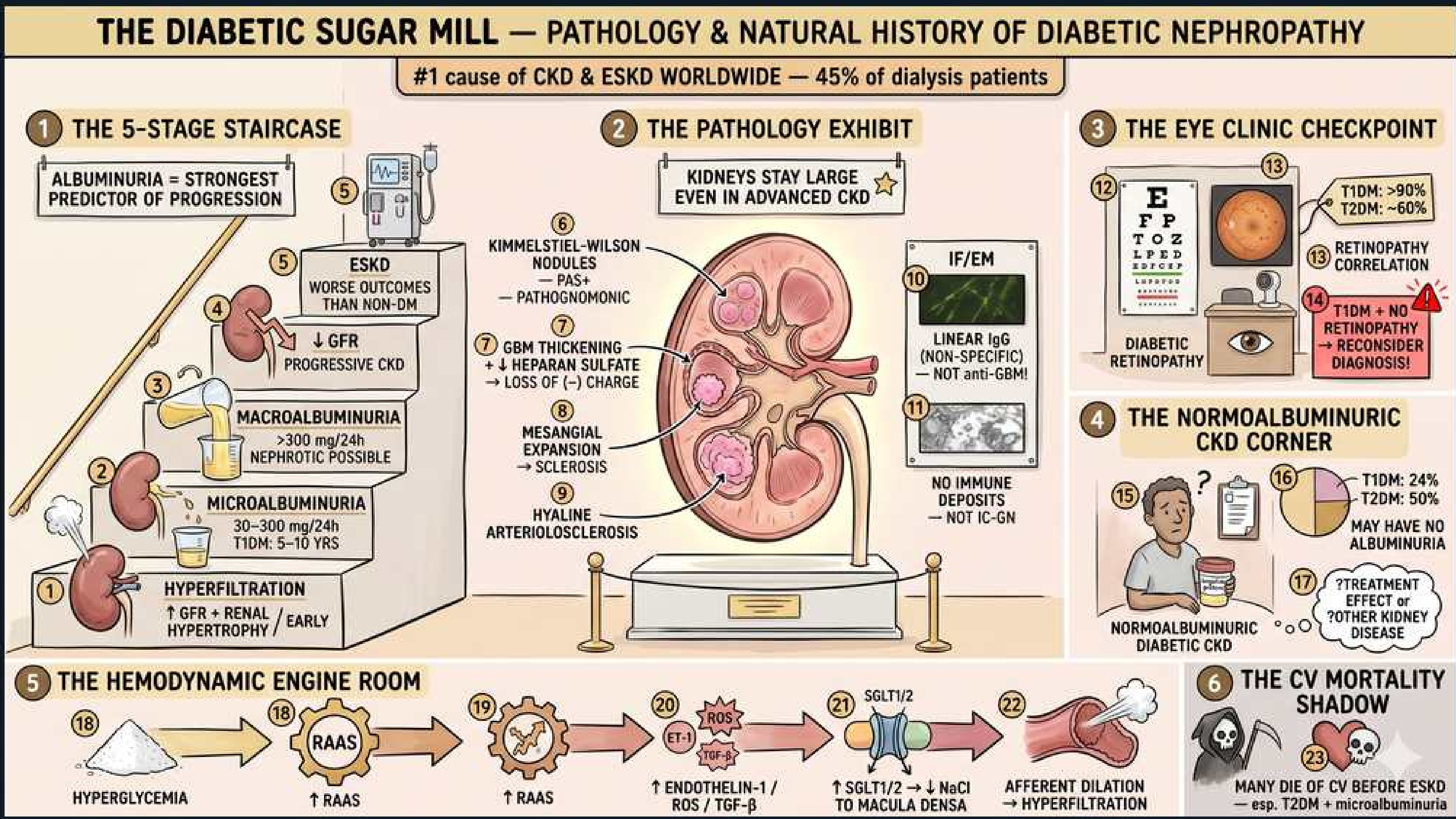
Diabetic Nephropathy — Pathology & Natural History



1. #1 cause of ESKD

WHAT YOU SEE

Banner: leading cause of CKD & ESKD, 45% of dialysis patients

WHAT IT ENCODES

Diabetic nephropathy is the single most common cause of CKD and ESKD worldwide and accounts for roughly 45% of patients on dialysis. It develops in ~30-40% of type 1 and type 2 diabetics. Screening is by annual urine albumin-to-creatinine ratio (UACR) and eGFR.

BOARD TRAP

FSGS and hypertensive nephrosclerosis are common but NOT the #1 cause of ESKD — diabetes is.

2. Albuminuria = strongest predictor

WHAT YOU SEE

Step 1 of staircase: albuminuria is strongest predictor of progression

WHAT IT ENCODES

Albuminuria is the earliest clinical marker and the strongest predictor of progression to ESKD. The classic natural history climbs a staircase: hyperfiltration → microalbuminuria → macroalbuminuria → declining GFR → ESKD.

BOARD TRAP

In classic diabetic nephropathy albuminuria precedes GFR decline; an early FALL in GFR with little proteinuria suggests non-diabetic kidney disease.

3. Hyperfiltration (earliest)

WHAT YOU SEE

Bottom step: hyperfiltration, increased GFR + renal size, very early

WHAT IT ENCODES

The earliest functional change is glomerular hyperfiltration with an elevated GFR and enlarged kidneys, driven by afferent arteriolar dilation and efferent constriction (RAAS/angiotensin II). This precedes any albuminuria.

BOARD TRAP

Early diabetic kidneys are LARGE with a HIGH GFR — small shrunken kidneys early argue against diabetic nephropathy.

4. Microalbuminuria 30–300

WHAT YOU SEE

Step: microalbuminuria 30-300 mg/24h, appears 5-10 yrs

WHAT IT ENCODES

Moderately increased albuminuria (30–300 mg/g, formerly microalbuminuria) typically appears 5–10 years into disease and signals early nephropathy. It is the trigger to start an ACE inhibitor or ARB regardless of blood pressure.

BOARD TRAP

Don't dismiss 'micro'-albuminuria as benign — it mandates RAAS blockade and predicts both renal and cardiovascular events.

5. Macroalbuminuria >300

WHAT YOU SEE

Step: macroalbuminuria >300 mg/24h, nephrotic possible

WHAT IT ENCODES

Severely increased albuminuria (>300 mg/g) can reach nephrotic-range proteinuria (>3.5 g/day). Diabetic nephropathy is one of the most common causes of nephrotic syndrome in adults.

BOARD TRAP

Nephrotic-range proteinuria in a long-standing diabetic with retinopathy is usually diabetic nephropathy — biopsy not always required.

6. GFR decline → ESKD

WHAT YOU SEE

Top steps: progressive CKD then ESKD, worse outcomes than non-DM

WHAT IT ENCODES

Once macroalbuminuria sets in, GFR declines progressively to ESKD. Diabetic ESKD patients have worse outcomes (more CV death) than non-diabetic ESKD patients.

BOARD TRAP

Strict glycemic and BP control slow but do not fully halt progression once macroalbuminuria is established.

7. Kimmelstiel–Wilson nodules

WHAT YOU SEE

Pathology exhibit: Kimmelstiel-Wilson nodules, PAS+, pathognomonic

WHAT IT ENCODES

Nodular glomerulosclerosis — Kimmelstiel–Wilson nodules — are PAS-positive acellular mesangial nodules and are the pathognomonic light-microscopy lesion of diabetic nephropathy. Mesangial expansion and diffuse sclerosis also occur.

BOARD TRAP

Kimmelstiel–Wilson nodules look similar to nodular sclerosis of light-chain deposition disease/amyloid — but KW is PAS+ and Congo-red NEGATIVE.

8. GBM thickening + heparan loss

WHAT YOU SEE

GBM thickening + heparan sulfate, loss of negative charge

WHAT IT ENCODES

Thickening of the glomerular basement membrane (earliest EM change) plus loss of negatively charged heparan sulfate reduces the charge barrier, permitting albumin leak. Mesangial expansion follows.

BOARD TRAP

GBM thickening on EM is the EARLIEST structural lesion, even before nodules appear on light microscopy.

9. Hyaline arteriolosclerosis

WHAT YOU SEE

Hyaline arteriolosclerosis, no immune deposits — not IC-GN

WHAT IT ENCODES

Diabetes causes hyaline arteriolosclerosis of BOTH afferent and efferent arterioles (vs hypertension, which affects mainly afferent). There are NO immune complex deposits — IF is essentially negative except nonspecific linear staining.

BOARD TRAP

Don't call diabetic nephropathy an immune-complex GN; deposits are absent, and linear IgG is nonspecific, not anti-GBM disease.

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10. Linear IgG ≠ anti-GBM

WHAT YOU SEE

IF/EM panel: linear IgG (nonspecific), NOT anti-GBM

WHAT IT ENCODES

IF can show nonspecific LINEAR IgG and albumin staining along the GBM due to protein trapping — this mimics anti-GBM disease but is NOT immune-mediated and lacks anti-GBM antibody or crescents.

BOARD TRAP

Linear IgG on IF in a diabetic = nonspecific trapping; true anti-GBM (Goodpasture) has crescents, hemoptysis, and circulating anti-GBM antibody.

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11. Retinopathy correlation

WHAT YOU SEE

Eye clinic: retinopathy >90% (T1DM), correlates with nephropathy

WHAT IT ENCODES

Diabetic retinopathy strongly correlates with nephropathy: present in >90% of T1DM and ~60% of T2DM with nephropathy. Its presence supports a diabetic etiology for proteinuria.

BOARD TRAP

Proteinuria WITHOUT retinopathy (especially T1DM) should raise suspicion for a non-diabetic glomerular disease and prompt biopsy.

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12. Normoalbuminuric CKD

WHAT YOU SEE

Normoalbuminuric diabetic CKD corner: 24-50% may have no albuminuria

WHAT IT ENCODES

A growing subset (24–50%, esp. T2DM) develops declining GFR WITHOUT albuminuria — normoalbuminuric diabetic kidney disease, reflecting more ischemic/tubulointerstitial injury. eGFR must be tracked independently of albuminuria.

BOARD TRAP

Absence of albuminuria does NOT exclude diabetic kidney disease; relying on UACR alone misses normoalbuminuric CKD.

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13. Hemodynamic engine (RAAS)

WHAT YOU SEE

Engine room: hyperglycemia → RAAS → efferent constriction → hyperfiltration

WHAT IT ENCODES

Hyperglycemia activates RAAS, endothelin-1, ROS and TGF-β, causing efferent arteriolar constriction and afferent dilation → raised intraglomerular pressure and hyperfiltration. SGLT2-mediated sodium reabsorption reduces macula densa NaCl, blunting tubuloglomerular feedback and worsening hyperfiltration.

BOARD TRAP

SGLT2 inhibitors restore tubuloglomerular feedback (afferent constriction) and are renoprotective — the opposite of the hyperfiltration that drives injury.

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14. CV mortality shadow

WHAT YOU SEE

CV mortality shadow: most die of CV disease before ESKD

WHAT IT ENCODES

Most patients with diabetic nephropathy die of cardiovascular disease before reaching ESKD. Even microalbuminuria is an independent marker of cardiovascular risk.

BOARD TRAP

The dominant cause of death in diabetic nephropathy is cardiovascular, not uremia — manage CV risk aggressively.
